L13

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Find the zeroes & poles

$$G(s) = \begin{bmatrix} 1/(s+1) & 1/s+2 \\ 2/s+2 & 1/s+1 \end{bmatrix} \xrightarrow{\text{minimal realization}} (A, B, C, D)$$

Gilbert's method (distinct poles) $\lambda_i \neq \lambda_j$

$$\boxed{(s+1)(s+2)}$$

$$= \frac{1}{s+1}\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{W_1} + \frac{1}{s+2}\underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}}_{W_2}$$

$-1 = \lambda_1$, $-2 = \lambda_2$

$$1/s^2-1 = \frac{A}{s+1} + \frac{B}{s-1}$$

Step: $G(s) = D + \sum_{i=1}^{k} \frac{W_i}{s-\lambda_i}$

$\lambda_i \rightarrow$ roots of LCM of den. of each element of $G(s)$

$$LCM = \{(s+1)(s+2)(s+2)(s+1)\}$$
$$= (s+1)(s+2)$$

$r_i := \text{rank } W_i$ $\qquad r_1 = 2$, $r_2 = 2$

$A \in \mathbb{R}^{n \times n}$ $\qquad n = \sum_{i=1}^{k} r_i$ $\qquad n = 4$

$\downarrow$

$A \in \mathbb{R}^{4\times4}$, $B \in \mathbb{R}^{4\times2}$, $C \in \mathbb{R}^{2\times4}$, $D \in \mathbb{R}^{2\times2}$

$D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Step 2:-

$$A = \begin{bmatrix} \lambda_1 & & & & & \\ & \lambda_1 & & & & \\ & & \ddots & & & \\ & & & \lambda_{r_1} & & \\ & & & & \lambda_2 & \\ & & & & & \ddots \\ & & & & & & \lambda_{r_2} \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$$B = \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_k \end{bmatrix} \qquad C = \begin{bmatrix} C_1 & C_2 & \cdots & C_k \end{bmatrix}$$

$$W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{C_1} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{B_1}$$

$$W_2 = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{C_2} \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}}_{B_2}$$

$W_1 = C_1 B_1$, $W_2 = C_2 B_2 \cdots W_k = C_k B_k$

$C_i \in \mathbb{R}^{p \times r_i}$, $B_i \in \mathbb{R}^{r_i \times m}$ $\qquad i \in \{1, 2, \ldots k\}$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \hline 0 & 1 \\ 2 & 0 \end{bmatrix} \qquad C = \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array}\right]$$

$$\text{rank} \begin{bmatrix} A - \sqrt{2}I & B \\ C & D \end{bmatrix} \overset{5}{<} \text{nrank} \begin{bmatrix} A - sI & B \\ C & D \end{bmatrix} \overset{6}{}$$

e.g. $G(s) = \begin{bmatrix} 1/s+1 & 2/s+1 \\ \frac{1}{s+2} - \frac{1}{s+1} & 1/s+2 \end{bmatrix}$ LCM $= (s+1)(s+2)$

$-1/(s+1)(s+2)$

$= \frac{1}{s+2} - \frac{1}{s+1}$

$$= \frac{W_1}{s+1} + \frac{W_2}{s+2}$$

$$G(s) = \frac{\begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}}{s+1} + \frac{\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}}{s+2}$$

$r_1 = 2$
$r_2 = 1$ $\quad n = 2+1 = 3 \quad A \in \mathbb{R}^{3\times 3}$

$\{-1, -1, -2\}$ $\quad A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 1 & 1 \end{bmatrix}$

$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$W_1 = C_1 B_1 \quad B_1 \in \mathbb{R}^{2\times 2}, \; C_1 \in \mathbb{R}^{2\times 2}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$

$W_2 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \quad B_2 \in \mathbb{R}^{1\times 2}, \; C_2 \in \mathbb{R}^{2\times 1}$

$= \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{C_2} \underbrace{\begin{bmatrix} 1 & 1 \end{bmatrix}}_{B_2}$

Transmission zeros:

Smith form of a polynomial matrix: If $P(s) \in \mathbb{R}^{p\times m}[s]$, $\exists\, U(s), V(s)$ s.t.

$$U(s)\, P(s)\, V(s) = \begin{bmatrix} \gamma_1(s) & & & & \\ & \gamma_2(s) & & & \\ & & \ddots & & \\ & & & \gamma_r(s) & \\ & & & & 0 \\ & & & & & \ddots \; 0 \end{bmatrix}, \text{ where } \gamma_i(s) \text{ divides } \gamma_{i+1}(s)$$

e.g. $P(s) = \begin{bmatrix} s+2 & s+1 \\ 2(s+1) & s+2 \end{bmatrix}$

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ s & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}}_{U(s)} \begin{bmatrix} s+2 & s+1 \\ 2(s+1) & s+2 \end{bmatrix} \underbrace{\begin{bmatrix} 1/2 & -1/2\, s \\ 0 & 1 \end{bmatrix}}_{V(s)} = \begin{bmatrix} 1 & 0 \\ 0 & s^2-2 \end{bmatrix}$$

McMillan form: If $G(s) \in \mathbb{R}^{p\times m}(s)$, then $\exists\, W(s), Q(s)$ such that

$$W(s)\,G(s)\,Q(s) = \begin{bmatrix} \frac{\alpha_1(s)}{\beta_1(s)} & & & & & \\ & \frac{\alpha_2(s)}{\beta_2(s)} & & & & \\ & & \ddots & & & \\ & & & \frac{\alpha_r(s)}{\beta_r(s)} & & \\ & & & & 0 & \\ & & & & & \ddots \\ & & & & & & 0 \end{bmatrix}, \quad \begin{array}{l} \alpha_i(s) \text{ divides } \alpha_{i+1}(s) \\ \beta_{i+1}(s) \text{ divides } \beta_i(s) \end{array}$$

e.g.

$$G(s) = \begin{bmatrix} 1/_{s+1} & 1/_{s+2} \\ 2/_{s+2} & 1/_{s+1} \end{bmatrix} = \frac{1}{(s+1)(s+2)} \underbrace{\begin{bmatrix} s+2 & s+1 \\ -2(s+1) & s+2 \end{bmatrix}}_{P(s)}$$

Smith form

$$W(s)\,G(s)\,Q(s) = \frac{1}{(s+1)(s+2)} \begin{bmatrix} 1 & 0 \\ 0 & s^2-2 \end{bmatrix} = \begin{bmatrix} 1/\underbrace{(s+1)(s+2)}_{\beta_1(s)} & 0 \\ 0 & (s^2-2)/\underbrace{(s+1)(s+2)}_{\beta_2(s)} \end{bmatrix}$$

- The number $\sum_{i=1} \deg \beta_i(s)$ is called $\underbrace{\text{McMillan degree}}_{\text{minimum number of states required to realize the system.}}$. e.g. 4
- The roots of $\beta_i(s)$ in McMillan form of $G(s)$ are the poles of the system.
  e.g. $\{-1, -2, -1, -2\}$
- The roots of $\alpha_i(s)$ in McMillan form of $G(s)$ are the transmission zeroes of the system.
  e.g. $\{\pm\sqrt{2}\}$
- If $G(z_0) = 0$, then z_0 is a blocking zero.

Given $G(s) \in \mathbb{R}^{p\times m}(s)$ (proper) and (A,B,C,D) being its minimal realization, z_0 is a transmisson zero of $G(s)$ if and only if z_0 is an invariant zero of (A,B,C,D)

Every TZ is an invariant zero of the system. $\quad TZ \subseteq I.Z$

e.g.
$$\left[\begin{array}{ccc|c} 1 & 4 & 0 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 2 & -3 & -1 \\ \hline -1 & -1 & 0 & 0 \end{array}\right]$$

What are T.Z and I.Z ?

$T.Z \neq I.Z$

$\sigma(A) = \{1, -1, -3\}$

T.Z : $G(s) = \dfrac{(s+3)\cancel{(s+3)}}{(s-1)(s+1)\cancel{(s+3)}}$ $\quad$ T.Z : $\boxed{\{-3\}}$ $\{1, -1\}$

I.Z :
$$M(s) := \begin{bmatrix} 1-s & 4 & 0 & 0 \\ 0 & -1-s & 0 & -1 \\ 0 & 2 & -3-s & -1 \\ -1 & -1 & 0 & 0 \end{bmatrix}$$

$\text{rank}(M(-3)) = 2$

$\text{nrank}(M(s)) = 4$

$I.Z. = \{\underbrace{-3}_{T.Z}, \underbrace{-3}_{\text{I.Z which is not T.Z}}\}$

Suppose z_0 is not a pole of $G(s)$. Then z_0 is T.Z $\iff$

$$\underbrace{\text{rank}(G(z_0))}_{\Downarrow} < \underbrace{\text{normal rank}(G(s))}$$

I.Z

$$\text{rank}\begin{bmatrix} A-z_0 I & B \\ C & D \end{bmatrix} < \text{nrank}\begin{bmatrix} A-sI & B \\ C & D \end{bmatrix}$$

$$G(s) = \begin{bmatrix} 1/s+1 & 1/s+2 \\ 2/s+2 & 1/s+1 \end{bmatrix} = \frac{1}{(s+1)(s+2)} \underbrace{\begin{bmatrix} s+2 & s+1 \\ 2(s+1) & s+2 \end{bmatrix}}_{P(s)}$$

$\text{nrank}(G(s)) = \text{nrank}(P(s))$

Lemma: Suppose $\begin{bmatrix} A-sI & B \\ C & D \end{bmatrix}$ has F.C. normal rank. Then z_0 is an I.Z of $(A,B,C,D) \iff \exists\, x \neq 0$ and $u \in \mathbb{C}^m$ such that

$$\begin{bmatrix} A-z_0 I & B \\ C & D \end{bmatrix} \begin{bmatrix} \boxed{x} \\ u \end{bmatrix} = 0 \overset{u=0}{\Rightarrow} \underbrace{\begin{bmatrix} A-z_0 I \\ C \end{bmatrix} x = 0}_{\boxed{PBH}\ *}$$

Moreover, if $u=0$, then z_0 is also unobservable.

e.g.

$$M(s) = \begin{bmatrix} 1-s & 4 & 0 & 0 \\ 0 & -1-s & 0 & -1 \\ 0 & 2 & -3-s & -1 \\ -1 & -1 & 0 & 0 \end{bmatrix} \qquad \ker\{M(-3)\} = \left\langle \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \\ -2 \end{pmatrix} \right\rangle$$

unobs.

$\{-3, -3\}$ observable

$\boxed{\text{Th}}$ $G(s) \in \mathbb{R}^{p\times m}(s)$ with full column normal rank. Then $z_0 \in \mathbb{C}$ is a T.Z of $G(s) \iff \exists\, u \neq 0$ s.t $G(z_0)\, u_0 = 0$

$*$

$$\begin{bmatrix} 1/s+1 & 1/s+2 \\ 2/s+2 & 1/s+1 \end{bmatrix} \qquad \ker\{G(-\sqrt{2})\} = \left\langle \begin{pmatrix} * \\ * \end{pmatrix} \right\rangle \quad 1 \text{ dim.}$$

e.g. $$G(s) = \begin{bmatrix} 1/s+1 & 1/s+1 \\ 1/s+1 & 1/s+1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0 \quad \forall\, s \in \mathbb{C}.$$

Lemma: Suppose $\begin{bmatrix} A-sI & B \\ C & D \end{bmatrix}$ has F.R. normal rank. Then z_0 is an I.Z of $(A,B,C,D) \iff \exists\, y \neq 0$ and $v \in \mathbb{C}^p$ such that

$$[y^* \quad v^*] \begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} = 0 \overset{v=0}{\Rightarrow} \underbrace{y^* [A - z_0 I \quad B] = 0}_{\text{PBH test}}$$

Moreover, if $v = 0$, then z_0 is also uncontrollable.

$\boxed{\text{Th}}$ $G(s) \in \mathbb{R}^{p\times m}(s)$ with full row normal rank. Then $z_0 \in \mathbb{C}$ is a T.Z of $G(s) \iff \exists\, \eta \neq 0$ s.t $\eta^* G(z_0) = 0$

$*$

$\boxed{\text{Th}}$ $G(s)$ is F.C.N.R (F.R.N.R) $\iff$ $\begin{bmatrix} A-sI & B \\ C & D \end{bmatrix}$ has F.C.N.R (F.R.N.R)

Lemma: Let $G(s) \in \mathbb{R}^{p \times m}(s)$ be a transfer matrix & (A, B, C, D) be a minimal realization. If the system i/p is of the form $u(t) = u_0 e^{\lambda t}$, where $\lambda \in \mathbb{C}$ is not a pole of $G(s)$ and $u_0 \in \mathbb{C}^m$ is an arbitrary vector, then o/p due to $u(t)$ and $x(0) = (\lambda I - A)^{-1} B u_0$ is

$$\boxed{y(t) = G(\lambda) e^{\lambda t} \quad \forall \; t \geq 0.}$$

In particular, if λ is a zero of $G(s)$ then $y(t) = 0$.